

Evaluation of a new bond-type anchorage system with expansive grout for a single FRP rod

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HIGHLIGHTS

- Pullout tests were conducted on twelve CFRP bond-type anchors and eight specimens with built-in anchors.
- The effects of bond length, rod-to-anchor diameter ratio, and curing time were investigated.
- The anchor length must not be less than 32- and 30-times rod diameter for #3 (10 mm) and #4 (13 mm) CFRP rods.
- Results obtained from numerical analysis are in good agreement with the experimental results.

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ABSTRACT

Prestressed concrete applications require a high strength prestressing material that could apply and maintain effective compressive force to concrete members. Although fiber reinforced polymer (FRP) materials, especially carbon FRP (CFRP), have the desired strength, no efficient system for its anchorage to concrete has been devised yet. This paper presents an experimental evaluation on a new bond-type anchorage system using an expansive grout as a filling material to anchor CFRP rods. A total of twelve specimens were fabricated and tested. Short-term and long-term pullout tests were performed. The CFRP rods were 0.375 in. (10 mm) and 0.50 in. (13 mm) in diameter. The parameters investigated were the bond length, rod diameter, and curing time of the expansive grout. The experimental results showed that 12 in. (305 mm) and 15 in. (381 mm) were adequate bond lengths to effectively anchor the 0.375 in. (10 mm) and 0.50 in. (13 mm) CFRP rods, respectively. The results also indicated that the anchor strength and stiffness were directly proportional to the bond length, and cross-sectional area ratio of CFRP rods to anchor borehole affected the stiffness and bonding capacity of the proposed anchor. Finally, a numerical analysis was carried out to predict the tensile behavior of FRP anchors. The numerical results had good agreement with the experimental results.

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1. Introduction

The use of fiber reinforced polymer (FRP) materials in civil engineering applications has become very common recently [1,2]. Researchers have proposed FRP as the most attractive materials to replace steel reinforcement in reinforced and prestressed concrete structures [3–6]. The attractive properties of FRPs include, but not limited to, high tensile strength, no corrosion, relatively

high modulus of elasticity, light weight, and suitable properties in terms of fatigue, creep and relaxation [7–12]. Among all types of FRP materials, carbon FRP (CFRP) is the most used materials in prestressed concrete applications due to its high tensile strength and modulus of elasticity [1,2,13]. However, a reliable anchorage system must be developed in order to efficiently utilize the superior properties of CFRP rods [14–18]. Because CFRP is significantly weaker in the transverse direction, the conventional anchorage systems used for steel strands are usually not suitable for CFRP rods as they may cause premature failure [1,15,19,20]. Tests have shown that the ultimate tensile capacity of CFRP rods used in prestressed concrete is controlled mainly by the anchorage capacity

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rather than the capacity of the rod itself [9,21,22]. Therefore, much research has been conducted to develop a simple and reliable FRP anchorage system [8,21–24]. The findings of the previous research resulted in several novel anchorage systems [10,13,20,22,25–30]. In fact, CFRP bridges have already been constructed in the United States and China [31,32]. However, research has shown that an FRP prestressing system that's competitive to steel prestressing system has not been devised yet, and more studies are required to develop a simple and reliable anchorage system for FRP rods or tendons [1,2,8,15,16].

In general, there are two types of FRP anchors available in the literature; mechanical anchors [22,27,29,33–37] and bond-type anchors [4,12,20,23,26,38,39]. The load carrying capacity of the mechanical anchors mostly depend on the friction between the FRP rods and the interior surface of the anchorage. In mechanical anchors, FRP rods must be subjected to a compressive force in the transverse direction (perpendicular to the longitudinal direction of the fibers). Since FRP is an anisotropic material, and weak in the transverse direction, FRP rod may fail due to compressive transverse forces before developing its maximum tensile strength. Nonetheless, mechanical anchors are commonly preferred because they are easier to install and require shorter anchorage length [8,34]. Unlike bond-type anchors, there is no curing time in mechanical anchors. The most common problems with mechanical anchors are the uncontrolled stress concentration, and the fact that the lateral forces meant to hold CFRP in place may penetrate the surface of FRP rod and cause premature failure [21,27,30]. On the other hand, a bond-type anchor mainly consists of a metal tube and grout holding the FRP rod inside. The anchorage strength mostly depends on the dimensions and properties of the metal tube, geometry and properties of FRP rod, and thickness and properties of the filling grout [2,17,38]. Research has shown that bond-type anchorage systems are the most reliable and efficient types of anchors for CFRP tendons [1,15]. Researchers have also indicated that bond-type anchors have favorable mechanical characteristics [17,20]. Bond-type anchorage systems tend to better distribute the lateral stresses on the surface of CFRP rods.

Based on its geometry, bond-type anchorage systems can be divided into three different types; straight anchors [40–42], conic anchors [43–47], and a combination between straight and conic anchors [48]. Although conic anchors have higher bond strength [4,13], the simplest bond-type anchorage system is the straight anchor. Several studies have been conducted to investigate the different parameters affecting the efficiency and bond strength of bond-type anchorage systems. In terms of the bonding material (grout), different materials have been tested. A cement-based material [12,20,42,46,47] or an epoxy adhesive [4,15,40,41,48] can be used as the filling grout. Cementitious materials are the most available and the cheapest material to be used. Benmokrane et al. [23] used different types of grout materials and conducted pullout tests on straight bond-type anchors. The authors concluded that grout containing “swelling agent” improved the bond strength. The swelling grout provided lateral pressure during curing and significantly improved the bond between the grout and FRP rods. The same observation was made by Zhang and Benmokrane [49]. Studies that focused on testing bond-type anchorage systems with expansive grout come mainly from two resources. Harada [50,51] is one of the first pioneers to develop a novel bond-type anchor using expansive grout to anchor CFRP rods. Harada's tests showed that #4 (13 mm) CFRP rod can be anchored to develop its full tensile strength by using a 10-in. (25 cm) bond-type anchor. The expansive pressure developed was 5.8 ksi (40 MPa) in 48 h. The second investigation about using expansive cement as the filling material was conducted by Lees et al. [52] on steel and aramid FRP rods. The authors used an expansive cement, commercially known as Bristar, that can develop a pres-

sure of 7.25 ksi (50 MPa) in 48 h. It was concluded that the expansive cement anchor could be used successfully (in laboratory-controlled conditions only) to anchor AFRP rods. No stress concentration was observed. Other than the above resources [50–52], information about using expansive grout as a filling material in bond-type CFRP anchorage systems is rarely available.

The main objective of the present study is to provide more information on the influence factors affecting the load-carrying capacity of CFRP bond-type anchorage systems using expansive cement as the bonding agent. The influence factors tested in this study were the bond length, the ratio of anchor to rod diameter, and curing time. Short and long-term pullout tests were conducted on twelve CFRP anchors made in the laboratory at Portland State University. Additional pullout tests were conducted on eight CFRP rods with built-in anchors obtained from the manufacturer. The results were analyzed, discussed, and compared with current acceptance standards. The results of this study provide additional information that may help current researchers develop a simple and reliable anchorage system for CFRP rods.

2. Experimental program

The testing program in this study was divided into two parts. The first part included tensile tests on the CFRP rods with built-in (BI) anchors that were already attached to the CFRP rods when purchased from the manufacturer. The tests were necessary to verify the tensile properties of the CFRP rods given by the manufacturer. In addition, the tests were used to evaluate whether the built-in anchoring system provided by the manufacturer could be used to develop the maximum tensile strength of the CFRP rods. All eight specimens with BI anchors were tested using a universal tensile test machine.

The second part of the testing program focused on evaluating the proposed bond-type (BT) anchorage system. Specimens with BT anchors did not fit in the universal tensile test machine. Therefore, a special test setup was made for specimens with BT anchors. On the other hand, it was difficult to test the specimens with built-in (BI) anchors in the special testing setup because the BI anchors were smooth and difficult to grip. As a result, specimens with BI anchors were tested using universal tensile test machine, and BT specimens were tested using a special test setup that will be described in section 2.3.3. Using two different test setups might affect the results slightly. However, the main goal of this study is not to compare the two types of anchors but to develop and test a proposed new bond-type (BT) anchorage system.

2.1. Material properties

2.1.1. CFRP rods

Aslan FRP rods, series Aslan 200 and Aslan 250, from Hughes Brothers, Inc. were used in this study. They were sand coated CFRP rods with helical wrap for bond improvement. The only difference between Aslan 200 and Aslan 250 series was that Aslan 250 came with a “proprietary factory applied anchorage” at the rods' ends. The tensile properties of the rods were determined and provided by the manufacturer following ASTM D7205 [53]. Table 1 presents the mechanical properties of CFRP rods. The tabulated ultimate tensile strength values are based on the definition of the “guaranteed” tensile strength provided by ACI 440 [54]. The modulus of elasticity is the slope of the stress–strain curve between 10% and 50% of the ultimate strength. The ultimate strain is estimated based on the ultimate load, nominal cross-sectional area, and the modulus of elasticity. In this study, the tensile properties of the CFRP rods were also determined experimentally using specimens made with built-in steel anchors (supplied by the manufacturer).

Table 1
Mechanical properties of CFRP reported by the manufacturer.

Strand size	Cross-sectional area		Guaranteed tensile strength		Ultimate strain	Tensile modulus of elasticity	
	in ²	mm ²	ksi	MPa		ksi	GPa
#3 (10 mm)	0.110	71.26	315	2172	1.75	18,000	124
#4 (13 mm)	0.196	126.68	310	2068	1.67	18,000	124

and the proposed bond-type anchors made in the laboratory. The experimental tests will be discussed in more details in the following sections.

2.1.2. Steel tube

The type of the steel tube that was used for fabricating the anchorage system in this study was originally manufactured to be used as a geo-drill injection anchor system for soil-anchorage applications. It is a continuously threaded steel tube with an inner diameter of 0.787 in. (20 mm), outer diameter of 1.25 in. (32 mm), and with a minimum net area through threads of 0.556 in² (360 mm²). The length of steel tube represents the anchorage bond length, or the anchor length. Three different lengths, 10 in. (254 mm), 12 in. (305 mm), and 15 in. (381 mm) were investigated in this study.

2.1.3. Filling grout

An expansive cementitious grout, commercially known as Bustar, was used as the filling material for the proposed bond-type anchor. This expansive grout is typically used for concrete and rock non-explosive demolition projects. In this study, it was used as a filling material to provide enough lateral pressure to hold the CFRP rod inside the steel tube. A water-to-cement (w/c) ratio of 30% was used even though a lower ratio would produce larger pressure based on the product data sheet. The reason for using 30% w/c ratio was to achieve reasonable workability and casting time. The mixing time was two to three minutes, and the casting time was within five minutes. According to the manufacturer, Bustar can develop an expansive pressure of at least 7.25 ksi (50 MPa) in 48 h.

2.2. Built-in (BI) anchors

A total of eight samples, four of #3 (10 mm) and four of #4 (13 mm) CFRP rods, were purchased from the manufacturer (Hughes Brothers, Inc). Information about the process used to attach the built-in (BI) anchors to CFRP rods was not available. The manufacturer named it “a proprietary factory applied anchorage” system that were pre-attached to both ends of the rods, as shown in Fig. 1. The total length, steel anchor dimensions, and CFRP free length of the samples with built-in (BI) steel anchors are presented in Table 2. The four samples in each size had the same dimensions. The total length of the samples was equal or less than 32 in. (813 mm) because that was the limit for the testing frame. In testing the samples, a hydraulic gripping system provided enough pressure to hold the smooth surface of the steel anchors from both ends (Fig. 1b). It was important to keep the eccentricity of the samples at minimum. The tensile tests were conducted following the guidelines provided by ASTM Standard [53]. The speed of testing was strain-controlled and was set to 0.005/minute. A 2-in. (51 mm) extensometer was used to monitor the strain during the test.

All eight samples were tested until failure. The tensile ultimate strength of the rods was calculated by dividing the measured peak load to the nominal cross-sectional area. The modulus of elasticity was determined from the stress–strain curve. To protect the extensometer, it was removed when the load reached 70% of the ultimate

guaranteed tensile capacity. As a result, the maximum strain was estimated from the peak load and the modulus of elasticity, assuming a linear stress–strain relationship until failure. Fig. 2 shows the measured stress–strain relationships compared to the manufacturer's values. In terms of the tensile modulus of elasticity, Fig. 2 shows that there is a good agreement between the guaranteed values from the manufacturer and the test results. Although the average maximum tensile strength for #4 (13 mm) rods was higher than the guaranteed strength, it was lower for #3 (10 mm) rods. The observed failure modes for #3 (10 mm) specimens with BI anchors were excessive slippage (BI#3-1 and BI#3-4), CFRP shear fracture at one of the interior edges of the BI anchors (BI#3-3), and rupture of helical CFRP at the anchor edge (BI#3-2). All #3 (10 mm) specimens with BI anchors failed at the anchorage zone. Therefore, the cause of failure was more likely due to the insufficient built-in (BI) steel anchors provided by the manufacturer. On the other hand, three of #4 (13 mm) specimens exceeded the guaranteed tensile capacity, and only one specimen (BI#4-4) failed to achieve the guaranteed tensile capacity. Specimens BI#4-1 and BI#4-4 failed due to excessive slippage, and specimens BI#4-2 and BI#4-3 failed due to CFRP rupture. The stress–strain relationships will be evaluated again while testing the proposed bond-type (BT) anchor.

2.3. Proposed bond-type (BT) anchors

A total of twelve samples were fabricated and tested in the laboratory. The goal of these tests was to evaluate the efficiency of the bond-type (BT) anchor and to check whether the proposed anchor could develop the full tensile strength of CFRP rods. The diameter of the CFRP rods were 0.375 in. (10 mm) and 0.50 in. (13 mm). The bond lengths, represented by the length of the steel tubes, were 10 in. (254 mm), 12 in. (305 mm), and 15 in. (381 mm).

2.3.1. Fabrication of the anchors

Fig. 3 shows the required items and the steps of fabricating the specimens. The steel tube was placed vertically in a wood frame (Fig. 3b). The steel tube sat on a PVC cap, which was supported by the wood frame. The PVC cap had a hole at the center where the CFRP rod went through. The cap was necessary to keep the CFRP rod centered and to keep the expansive grout from leaking. A predefined weight of the expansive grout and water (water-to-cement ratio = 0.3) was placed in a metal bowl (Fig. 3a). It was the preferred method in this study to add the water to the grout in two to three stages. This method made it easier to mix the sample and less likely to lose some drops of water during mixing. The mixing time was two to three minutes until the mixture became very soft (Fig. 3c–d). Then, the steel tube was filled with the water-grout mixture (Fig. 3e). After filling the steel tube with the grout, the CFRP rod was inserted inside the steel tube through the hole of the PVC cap to ensure that the CFRP rod was placed in the center. As mentioned before, the PVC cap at the bottom of the steel tube (as shown in Fig. 3a, b, g) had a hole at the center. At the top of the steel tube, a plastic cap with a hole in the middle was used to ensure that the CFRP rod was at the center on the top as well (Fig. 3f–g). Duct tape was used to keep the caps on top and



Fig. 1. (a) CFRP samples with built-in steel anchors, and (b) test set-up for the CFRP samples with built-in steel anchors.

Table 2
Typical dimensional details of the CFRP rods with built-in (BI) steel anchors obtained from the manufacturer.

Samples based on the size of CFRP (4 samples in each size)	Total length, in. (mm)	Steel anchor		Free length between anchors, in. (mm)
		Length, in. (mm)	Diameter, in. (mm)	
BI#3 (10 mm) CFRP rods	31.25 (794)	8.75 (222)	1.00 (25)	13.75 (349)
BI#4 (13 mm) CFRP rods	32.00 (813)	10.75 (273)	1.25 (32)	10.5 (267)

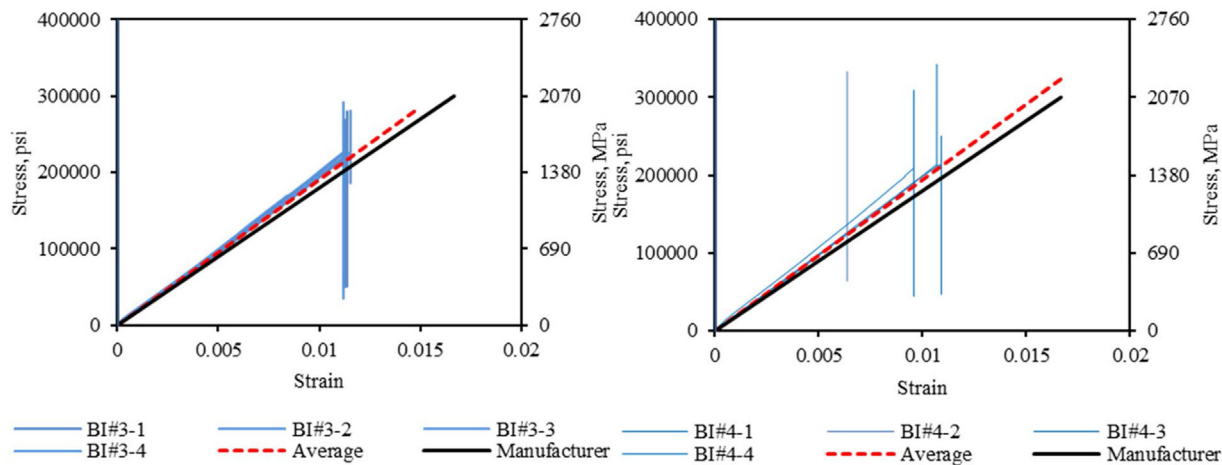


Fig. 2. Measured vs manufacturer's stress-strain relationships for #3 (10 mm) and #4 (13 mm) CFRP rods.

bottom in place. Since CFRP rods were centered from the top and bottom, they were most likely centered throughout the length of the steel tube. When the CFRP rod was inserted in the steel tube, enough grout spilled out indicating that little or no voids were left inside. Additionally, the nature of the expansive grout helped fill the voids, if any, by expanding its volume. The samples were then left for curing.

2.3.2. Specimen design

Two bond lengths (the length of the steel tube) for each size of CFRP rods were investigated. For #3 (10 mm) rods, 10 in. (254 mm) and 12 in. (305 mm) bond lengths were chosen. On the other hand, 12 in. (305 mm) and 15 in. (381 mm) bond lengths were chosen for #4 CFRP rods (Table 3). The other end (dead end) of all specimens were anchored with 20 in. (500 mm) steel anchors, and it will be called “the fixed end” throughout this paper. The free length of the CFRP rod between the two steel tubes was 9 in. (225 mm)

for all specimens. Approximately 5 in. (125 mm) to 10 in. (254 mm) of the rod were left at the dead end to hold a linear variable differential transformer (LVDT) that was used for slippage measurements. Fig. 4 shows dimensions and details of a typical specimen. Since the tube inner diameter was consistent, two different rod-to-anchor cross-sectional area ratios were investigated (Table 3). The last parameter was the curing time. All specimens were tested after four weeks of curing except two specimens (specimens BT#3-1 and BT#3-2) which were tested after one week of curing. The curing was applied by leaving the specimens in room temperature, 68° F (20° C).

2.3.3. Test setup and instrumentations

The test setup and instrumentation are presented in Fig. 5. Thick steel plates, 0.5 in. (13 mm) to 1.5 in. (38 mm), were welded longitudinally and transversely to create the steel box (testing frame) shown in Fig. 5. The two transverse plates had a hole at

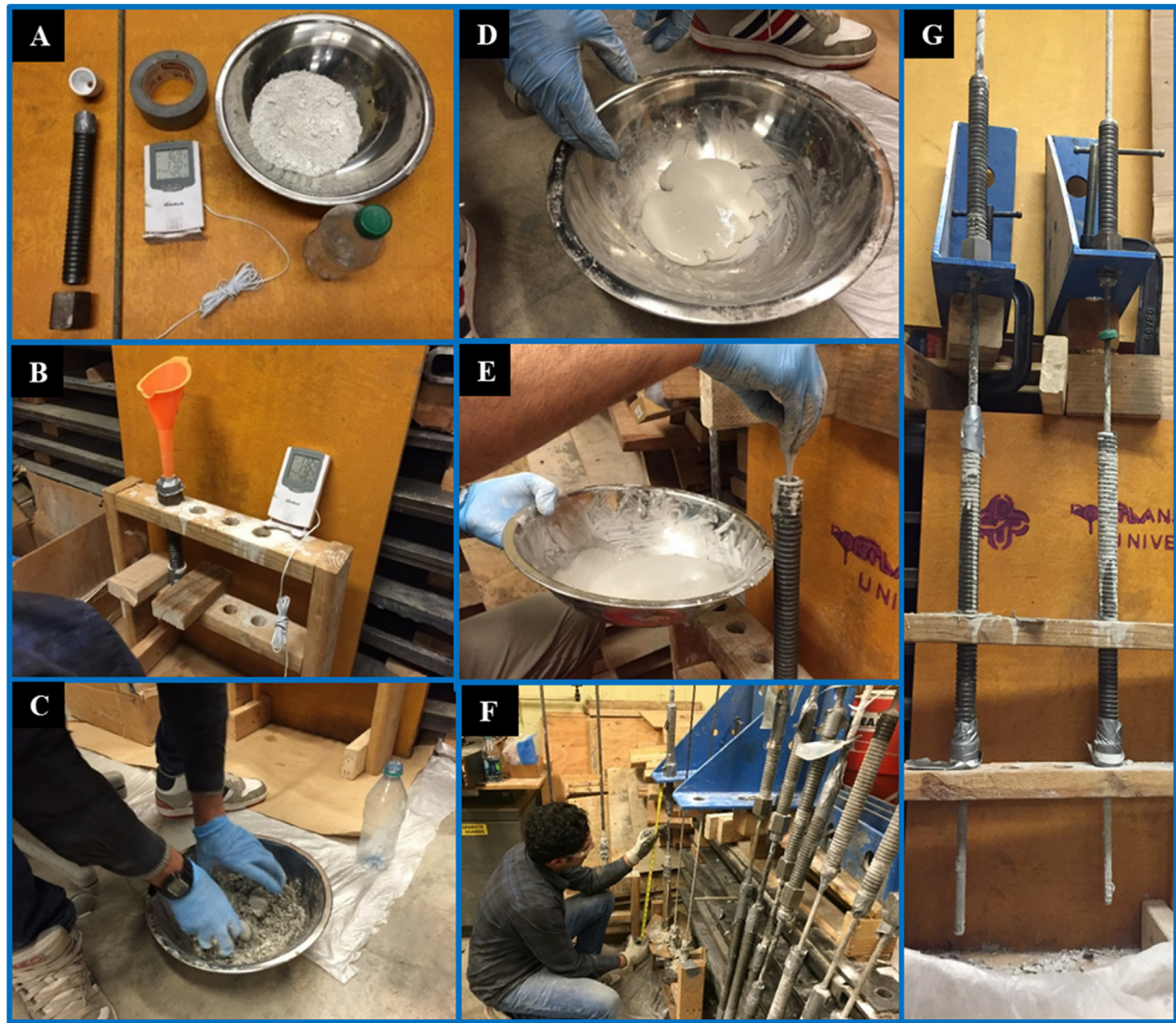


Fig. 3. Fabrication of CFRP bond-type anchor – (a): required items (listed from left to right, top to bottom): plastic PVC cap, steel tube, steel nut, CFRP rod, duct tape, thermometer, pan with expansive cement, and water, (b): wood frame to hold the steel tube vertically, (c): mixing the expansive cement with water, (d): expansive cement mixed with water, (e): pour expansive grout in the tube, and (f) and (g): alignment and curing of the anchors.

Table 3
Details of CFRP rods with bond-type (BT) anchors fabricated in the laboratory.

Specimen	Bond length*, in. (mm)	CFRP diameter, in. (mm)	Rod-to-anchor area ratio, %	Curing time
BT#3-1	10 (254)	0.375 (10)	22.7	4 days
BT#3-2	10 (254)	0.375 (10)	22.7	4 days
BT#3-3	12 (305)	0.375 (10)	22.7	4 weeks
BT#3-4	12 (305)	0.375 (10)	22.7	4 weeks
BT#3-5	12 (305)	0.375 (10)	22.7	4 weeks
BT#3-6	10 (254)	0.375 (10)	22.7	4 weeks
BT#4-1	12 (305)	0.500 (13)	40.4	4 weeks
BT#4-2	12 (305)	0.500 (13)	40.4	4 weeks
BT#4-3	15 (381)	0.500 (13)	40.4	4 weeks
BT#4-4	12 (305)	0.500 (13)	40.4	4 weeks
BT#4-5	15 (381)	0.500 (13)	40.4	4 weeks
BT#4-6	12 (305)	0.500 (13)	40.4	4 weeks

*Bond length is equal to the length of steel tube.

the center allowing the steel tubes (anchors) to go through from the hollow-core ram to the locking ends. The hollow-core ram was placed to push against a 1.5-in. (38 mm) thick plate, which transferred the load to the two longitudinal plates. The system allowed for locking the prestressing load using washers and hex

nuts (Fig. 5). The load was applied using a 10 ksi (70 MPa) hollow-core ram. The ram was connected to a hydraulic pump. The load was recorded using a strain gauge-based load cell. The axial strain was measured using sensitive strain gauges mounted on the surface of CFRP rods. One strain gauge was used per speci-

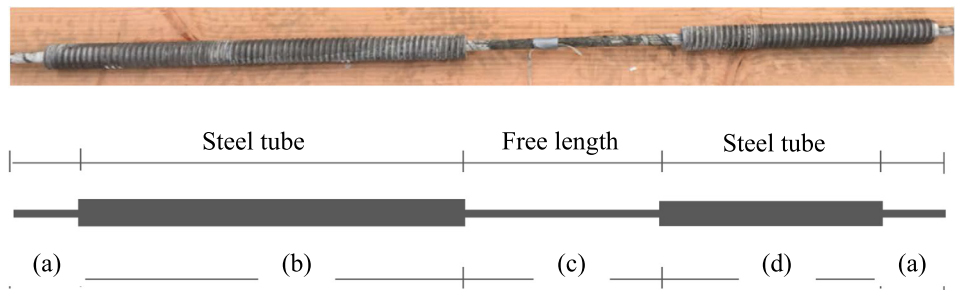
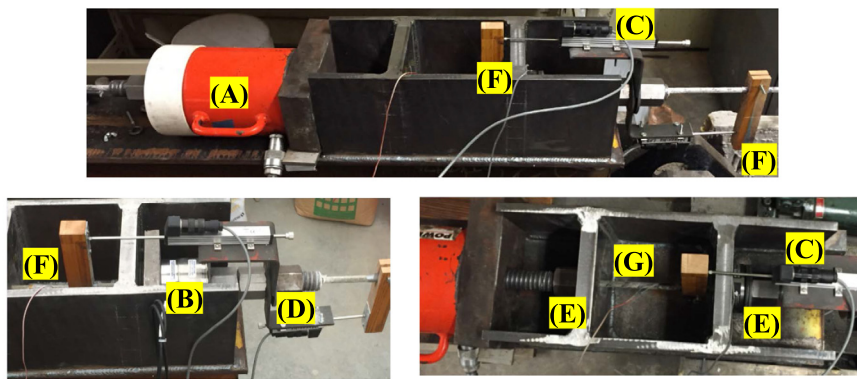
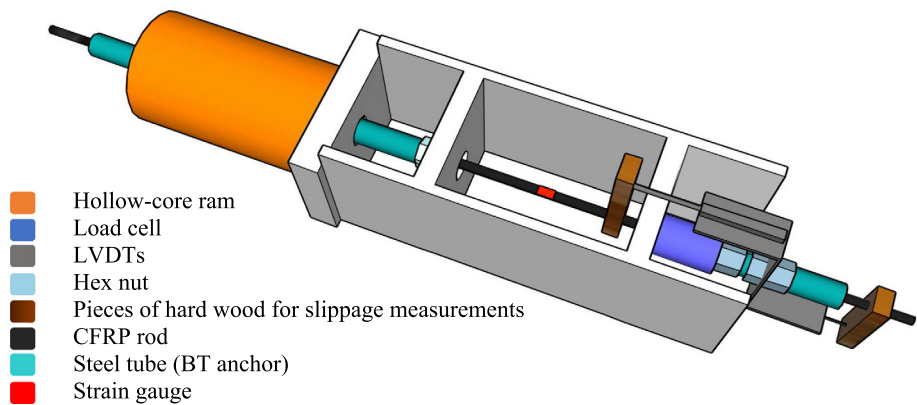


Fig. 4. Typical dimensions and details of the samples; (a): free length of CFRP at the dead end (5 in. [127 mm] to 10 in. [254 mm]), (b): fixed end anchor (20 in. [508 mm]), (c): free length of CFRP (9 in. [229 mm]), and (d): focused area; bond-type anchor (10 in. [254 mm], 12 in. [305 mm], and 15 in. [381 mm]).



(a) Pictures; (A): hollow-core hydraulic ram, (B): load cell, (C): LVDT-1, (D): LVDT-2, (E): steel washers and hex nut, (F): piece of hard wood, and (G): CFRP rod.



(b) Schematic drawing showing all parts

Fig. 5. Test setup and instrumentations.

men, as shown in Fig. 6. LVDT-2 was used to measure the end slip (dead end), while LVDT-1 was used to measure the slippage where the force was applied (loading end), as shown in Fig. 6. The applied force, axial strain, and displacements were monitored and collected using data loggers at a speed rate of twenty readings per second.

2.3.4. Loading procedure

Several guidelines and recommendations about testing FRP anchorage systems are available in the literature [55–57]. For the short-term tests, the loading procedure proposed by Schmidt et al. [22] found to be the most appropriate in this study. A load

rate of 1.0 mm/min was used. For the cyclic loading test, the cyclic load was increased gradually to 10%, 20%, 30%, 40%, 50%, 60%, and locked at 70% of the guaranteed tensile strength. The sustained loading test started immediately after the cyclic test. The pre-stressing force, axial strain, and displacements were monitored and recorded for ten days.

3. Results and discussion

In this section, the results of the experimental tests will be presented and discussed. The parameters of this study are the bond length and the rod diameter. The diameter of CFRP rods results

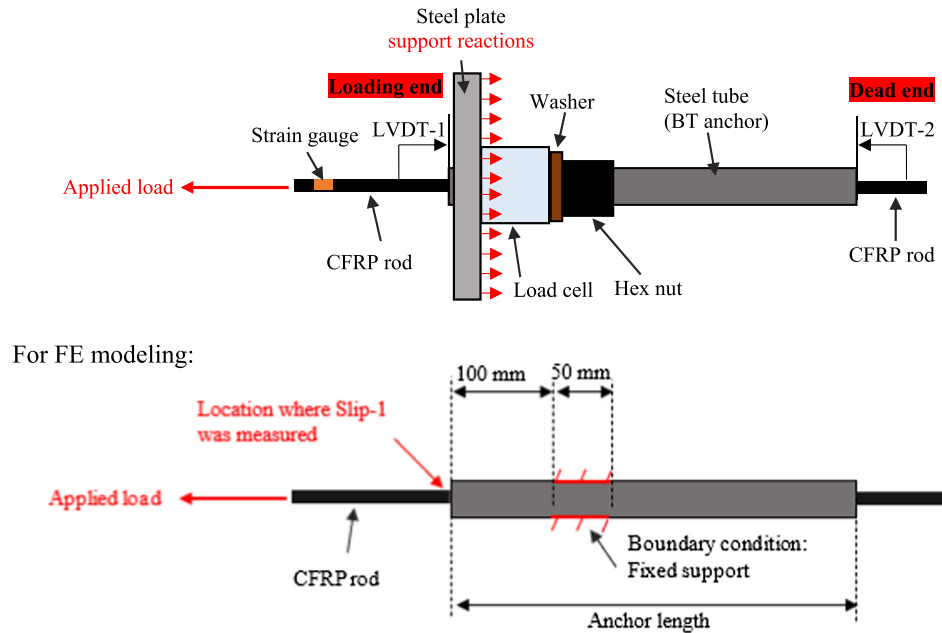


Fig. 6. Test setup and instrumentations; schematic drawing.

in different cross-sectional area ratio of the rod to anchor borehole. Furthermore, three specimens were tested to investigate the effects of curing time on the anchorage performance. All the samples were fabricated with steel tubes that had the same diameter. Each tube was a continuously threaded steel tube on the outside but smooth on the inside. The inner cross-sectional area of the tube was 0.486 in² (314 mm²). Therefore, the ratios of the cross-sectional area of CFRP rod to borehole were 22.7% and 40.4% for #3 (10 mm) and #4 (13 mm) rods, respectively. As shown in Fig. 6, the location where LVDT-1 measured displacement was called *loading end*, and the location where LVDT-2 measured displacement was called *dead end*. The slippage was measured at both locations (loading end and dead end) for all specimens, except for a few specimens that lost the readings of LVDT-2 due to instrumentation errors. The failure modes, stress-strain relationships of CFRP rods, CFRP mechanical properties, load vs slippage responses, effects of bond length, effects of cross-sectional area ratio, and effects of curing time will be discussed in the following sections.

3.1. Failure modes

There are two criteria for a successful anchor; (1) anchors should prevent FRP rupture to occur inside the anchorage, and (2) it must develop a minimum of 95% of the guaranteed tensile capacity of FRP rod [27,29]. Fig. 7 shows the modes of failure for #3 (10 mm) CFRP rods with bond-type (BT) anchors. All specimens failed mainly due to excessive slippage. In addition to slippage failure, rupture at the helical CFRP wrap was observed in specimen BT#3-4. This could be related to the quality of the rod used for specimen BT#3-4. On the other hand, specimen BT#3-6 failed due to FRP shear fracture at the interior edge of the anchor after a measured slippage of 3.25 mm. Stress concentration is one of the common reasons causing such a failure due to the weak properties of CFRP in transverse direction [27]. Quality of work and manmade errors during fabrication could be the reason of obtaining different results for specimens that have the same parameters. However, it is very difficult to consider the quality of work as one of the parameters. Therefore, it was assumed that all bond-type

(BT) specimens were fabricated with minimum variation in the quality of work.

Specimens BT#3-4 and BT#3-6 experienced the smallest slippage (2.5 mm and 3.25 mm, respectively), and had the most brittle failure modes. The slippage continued to increase for the other specimens for #3 (10 mm) rods until the test stopped. None of the 10 in. (254 mm) anchors could achieve an efficiency of 95% for #3 (10 mm) CFRP anchors. However, the efficiency for 12 in. (305 mm) anchors was at least 103%. Fig. 8 shows the modes of failure for #4 (13 mm) CFRP anchors. A sudden-brittle failure was the dominant behavior for these specimens, except specimen BT#4-2 which failed due to slippage only. The test stopped when the slippage exceeded 10 mm. Specimen BT#4-1 failed due to helical wrap rupture at the anchor edge. Due to stress concentration at the anchorage edge, rupture of CFRP helical wrap that confine the longitudinal fibers of the CFRP rod in a spiral shape was observed as shown in Fig. 8. CFRP rupture was the failure mode for specimens BT#4-3, BT#4-4, and BT#4-5. The failure mode for 15 in. (381 mm) anchors was more brittle than the failure mode for 12 in. (381 mm) anchors because the longer anchors failed in CFRP rupture. The efficiency of 15 in. (381 mm) anchors was more than 95%, and the failure occurred outside the anchorage. Therefore, these two anchors (BT#4-3 and BT#4-5) passed the both simple criteria for a successful anchor [27,29].

3.2. Tensile behavior of CFRP rods

Even though the main purpose of this study was to investigate the behavior of the proposed bond-type anchorage system, these tests provided important data regarding tensile properties of CFRP rods. In addition, some of the tensile tests that were performed on the CFRP rods with pre-attached built-in (BI) anchors from the manufacturer failed before reaching the ultimate tensile strength. As a result, the anchorage system investigation is also used to verify the tensile properties of CFRP rods. Figs. 9 and 10 show the measured stress-strain relationships for the tested CFRP rods. The strain of the CFRP rods was measured by strain gauges. Most of the time, the strain gauge readings were lost before the end of the tests mainly because the measured strain exceeded the reading

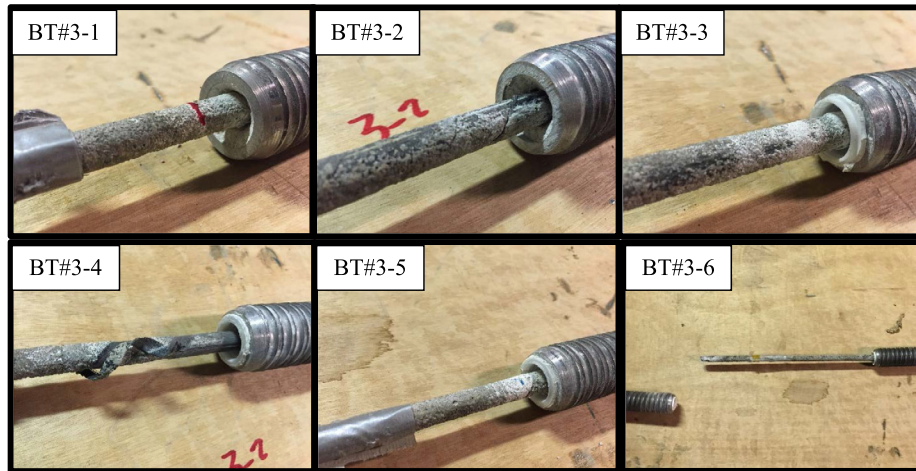


Fig. 7. Modes of failure for #3 (10 mm) CFRP anchors; all anchors failed in excessive slippage except BT#3-4 and BT#3-6 that failed due to rupture of helical CFRP.

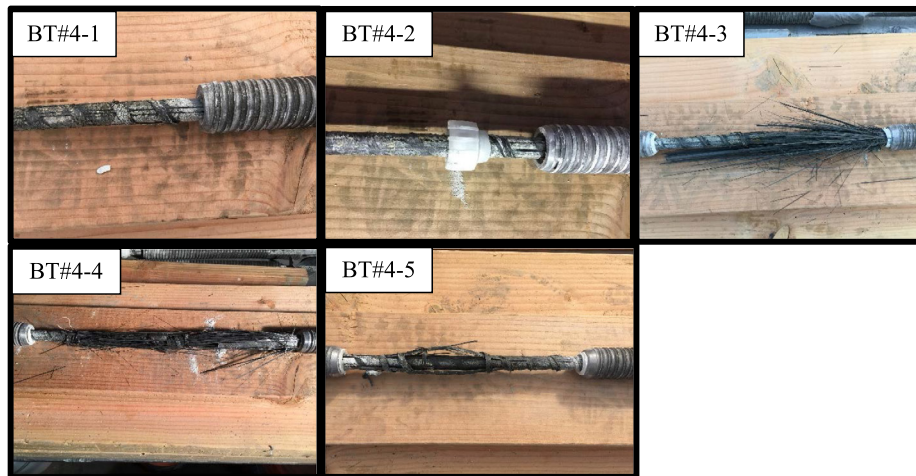


Fig. 8. Modes of failure for #4 (13 mm) CFRP anchors; all anchors failed in rupture except BT#4-1 (rupture of helical CFRP) and BT#4-2 (excessive slippage).

limit of the strain gauges. That was the reason why the experimental stress–strain relationships (Figs. 9 and 10) showed constant strain value at the end of the response while the measured load was increasing. The results of the tests are summarized in Table 4.

As shown in Table 4, none of #3 (10 mm) CFRP rods anchored with built-in (BI) anchors by the manufacturer (BI#3-1 through BI#3-4) achieved the guaranteed tensile capacity. Furthermore, the samples made with 10 in. (254 mm) bond-type anchors (BT#3-1, BT#3-2, and BT#3-6) failed to achieve the guaranteed tensile capacity as well. On the other hand, all samples made with 12 in. (305 mm) bond-type anchors (specimens BT#3-3, BT#3-4, and BT#3-5) were able to achieve at least 3% higher than the guaranteed tensile capacity of #3 (10 mm) CFRP rods. If all the ten tested specimens are considered, the mean value for tensile strength of #3 (10 mm) CFRP rod is 297 ksi (2,048 MPa), which is only 94% of the guaranteed capacity. However, if only the specimens with 12 in. (305 mm) bond-type anchors (specimens BT#3-3, BT#3-4, and BT#3-5) are considered, the mean value of the tensile strength is 346 ksi (2,388 MPa), which is 10% higher than the guaranteed capacity. Therefore, neither the pre-attached built-in (BI) anchors from the manufacturer, nor do the proposed bond-type (BT) anchors shorter than 12 in (305 mm) can be used to evaluate the tensile properties of #3 (10 mm) CFRP rods. In order to precisely evaluate the tensile strength of #3 (10 mm) CFRP rods,

specimens with 12 in. (13 mm) bond-type anchors or longer must be used.

The manufacturer claimed that they used a universal tensile test machine for testing based on ASTM Standards [53] similar to the one used in the present study, and their BI anchorage could be used to achieve the guaranteed tensile strength of #3 (10 mm) CFRP rods. It is possible that the BI anchors in the four specimens for #3 (10 mm) rods supplied by the manufacturer for this study were shorter than required to achieve the guaranteed tensile strength of #3 (10 mm) CFRP rods.

For #4 (13 mm) CFRP rods, all specimens achieved the guaranteed tensile strength, 300 ksi (2,068 MPa), except specimen BI#4-4, which was anchored with built-in (BI) steel anchors from the manufacturer. The range of the achieved tensile strengths was at least 3% and at most 30% higher than the guaranteed tensile strength of #4 (13 mm) CFRP rods, excluding specimen BI#4-4. The mean value of the tensile strength, standard deviation, and coefficient of variation are 339 ksi (2,339 MPa), 25 ksi (172 MPa), and 7.4%, respectively. The mean value is 13% higher than the guaranteed capacity.

If a guaranteed tensile strength were to be proposed based on the tested specimens following the definition of ACI 440 [54], *guaranteed tensile strength* = *mean value* – $3 \times$ *standard deviation*, the guaranteed tensile strength of #4 (13 mm) CFRP rods would be

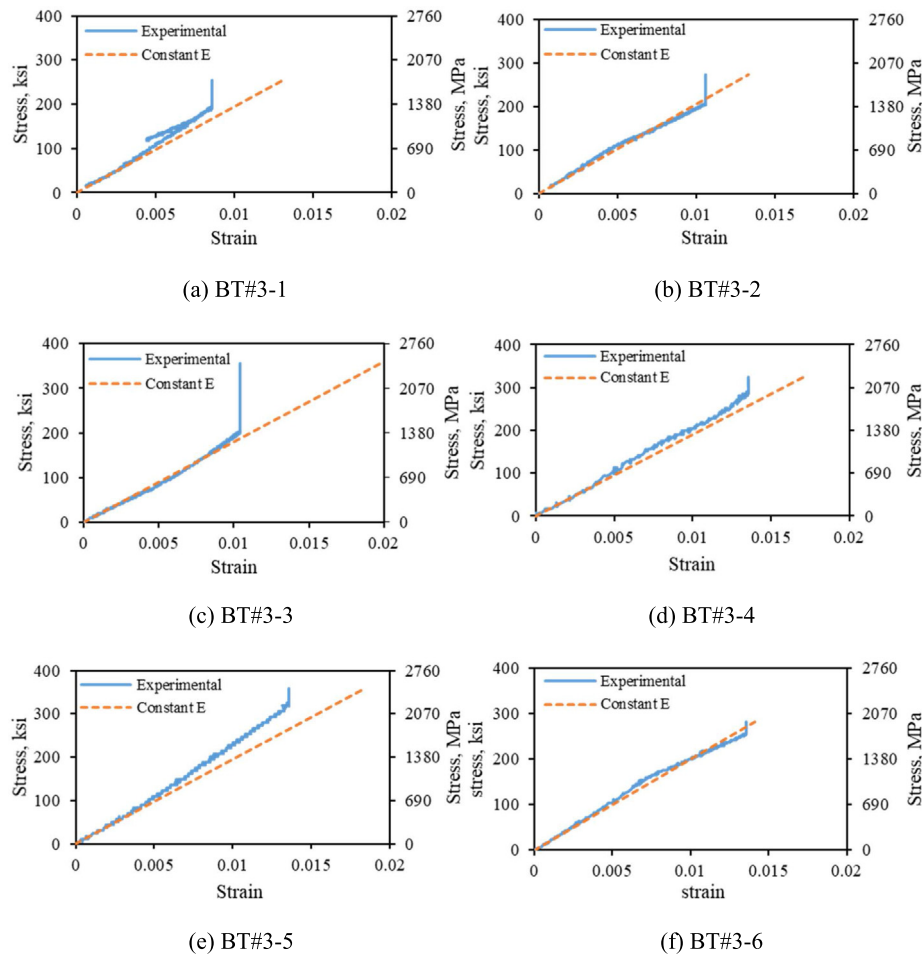


Fig. 9. Stress-strain relationships for #3 (10 mm) CFRP rods tested with proposed bond-type anchors.

264 ksi (1,822 MPa). Although it is safer to use the method of the guaranteed capacity proposed by ACI 440, it might be more reasonable to consider the 5-percent exclusion value from the frequency distribution curve of the tested results to determine the tensile capacity of CFRP rods. Based on the tested samples for #4 (13 mm) CFRP rods, the 5-percent exclusion value would be 298 ksi (2056 MPa). It is acknowledged, though, that the small number of specimens tested do not warrant a reliable statistical analysis.

The measured modulus of elasticity is the initial slope of the stress-strain relationship of CFRP rods. All measured values were equal to or exceeded the modulus of elasticity reported by the manufacturer. For #3 (10 mm) rods, the mean tensile modulus of elasticity was 19,280 ksi (133 GPa), but it was 19,730 ksi (136 GPa) for #4 (13 mm) CFRP rods. The range of the measured values of the tensile modulus was from 0.0% to 17% higher than the average modulus of elasticity reported by the manufacturer. It should be noted that the different test setups and loading rates for BI and BT anchors could have affected the results slightly. Table 4 shows that there was no significant difference in the results reported from the two different tests. For example, the averages of the reported moduli were 19,075 (131 MPa) for #3 (10 mm) rods with BI anchors, and 19,417 (134 MPa) for #3 (10 mm) rods with BT anchors.

3.3. Load vs Slip relationship

The initial slope of the load-slip relationship is the anchorage stiffness, which can be affected by many factors. These factors

include but not limited to (1) the bond length, (2) cross-sectional area ratio between the CFRP rod and anchor borehole, and (3) curing time for the cementitious grout. The slippage is recorded at the loading end and dead end of the anchor using two LVDTs, as shown in Figs. 5 and 6. The Load vs slippage curves are presented in Figs. 11 and 12. The ordinates of Figs. 11 and 12 represent the recorded load divided by the guaranteed capacity of CFRP strands, 315 ksi (2172 MPa) and 300 ksi (2068 MPa) for #3 (10 mm) and #4 (13 mm) rods, respectively. Slip-1 means the slippage recorded at the loading end, and slip-2 means the slippage recorded at the dead end. As expected, there was no slippage shown at the dead end (slip-2) until the tensile load was very close to the failure load. It is also shown that the slippage at the dead end follows the same pattern as the slippage at the loading end. Slippage at the dead end for anchors BT#4-3 and BT#4-5 was not recorded. The minimum length to develop the guaranteed strength for #3 (10 mm) and #4 (13 mm) CFRP rods was 12-inch (305 mm).

It is noteworthy that the decreasing of slip-1 in Fig. 11(c) occurred after a complete failure was observed and after the test was stopped. The time period between the maximum displacement of slip-1 (slip-1 = 9.75 mm and $P = 0.8 P_u$) and the point where the force was completely released (slip-1 = 7.5 mm and $P = 0$ kN) was fraction of a second. Therefore, the readings of LVDT-1 (Fig. 6) during this very short period of time did not represent decreasing of slip-1 during the test but more likely an instrumental error or a jump in the stroke of LVDT-1 because of the sudden drop of the applied load. This error or misrepresentation happened only with BT#3-3, and it is believed that it occurred

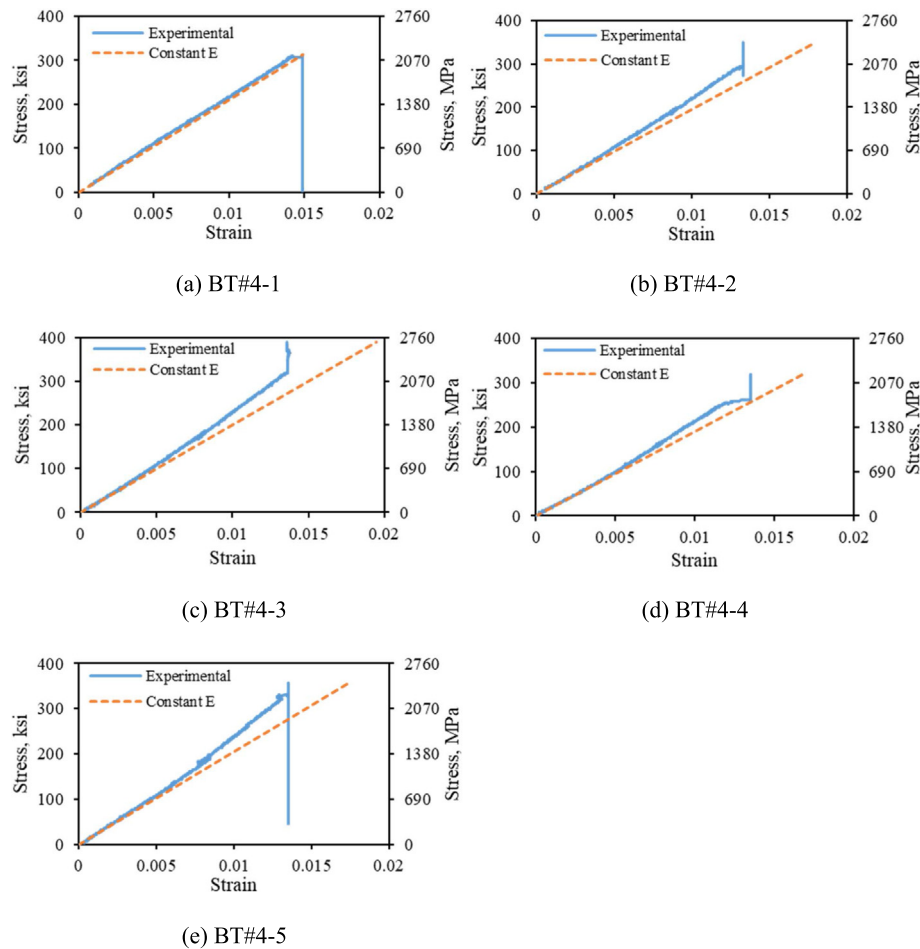


Fig. 10. Stress-strain relationships for #4 (13 mm) CFRP rods tested with proposed bond-type anchors.

due to inappropriate setup during the process of force releasing for this particular specimen.

3.4. Effects of bond length

The bond length is the length where the CFRP rod is in contact with the cementitious expansive grout inside the steel tube. Herein, it is the same length as the length of the steel tube. An anchor must have a minimum length in order to effectively distribute the bond stresses caused by the tensile load applied to the CFRP rod. If the bond stress is assumed to be uniformly distributed along the anchor length, the average grout-to-CFRP bond stress that should be resisted in order to develop the guaranteed capacity of CFRP rod can be calculated as the following:

$$Avg. bond stress at guaranteed load = \frac{Maximum guaranteed load}{Surface area of contact between CFRP and the Grout}$$

$$Avg. bond stress at guaranteed load = \frac{(Area of CFRP rod) \times (Max. guaranteed stress)}{\pi (perimeter of CFRP rod) \times (bond length)} \quad (4-1)$$

The average bond stress is either based on the assumption that the bond stresses are uniformly distributed along the bond length or based on the triangular stress distribution as described by ACI 440 [58]. Both assumptions lead to the same method of calculating the average bond stress (by using Eq.1). For #3 (10 mm) rods, the

anchors used were 10 in. (254 mm) and 12 in. (305 mm) long. Therefore, the average bond stresses for the 10 in. (254 mm) and 12 in. (305 mm) anchors were 2.95 ksi (20 MPa) and 2.46 ksi (17 MPa), respectively. The same procedure can be followed for #4 (13 mm) rods, and the average bond stresses for the 12 in. (305 mm) and 15 in. (381 mm) anchors were 3.12 ksi (22 MPa) and 2.49 ksi (17 MPa), respectively.

Considering #3 (10 mm) rods, Fig. 13 shows that the 10-inch (254 mm) anchors could achieve an average of 85% of the CFRP maximum guaranteed capacity. Five out of six specimens failed due to excessive slippage (Fig. 7). By using Eq. (4-1), 85% of the guaranteed strength generates an average bonding stress of 2.50 ksi (17 MPa) for 10 in. (254 mm) anchors. This value is very close to the average bond stress needed for 12-inch (305 mm) anchors to develop the maximum guaranteed capacity for #3 (10 mm) rods. If 2.50 ksi (17 MPa) is assumed to be the maximum bonding stress that can be handled for #3 (10 mm) rods, ignoring the effects of other factors and considering only the same rod size, the minimum bond length required to develop the maximum capacity of #3 (10 mm) CFRP rod is 11.8 in. (295 mm). This finding agrees with the results of 12-in. (305 mm) anchors. The measured tensile capacity of the 12-in (305 mm) anchors slightly exceeded the CFRP guaranteed strength. As it was observed in previous studies [15,42], anchorage bond length significantly affected the tensile strength of bond-type anchorage systems.

In the case of #4 (13 mm) CFRP rods, all the 12-in. (305 mm) and 15-in. (381 mm) anchors achieved the guaranteed capacity of #4 (13 mm) CFRP rods (Fig. 13). However, the average bond

Table 4

Summary of tensile test results.

Specimen	E_{measured} , ksi (GPa)	$E_{\text{manufacturer}}$, ksi (GPa)	$E_{\text{measured}} / E_{\text{manufacturer}}$	P_{max} , kip (kN)	$P_{\text{manufacturer}}$, kip (kN)	F_{max} , ksi (MPa)	$F_{\text{manufacturer}}$, ksi (MPa)	$F_{\text{max}} / F_{\text{manufacturer}}$	Failure modes
#3 (10 mm) CFRP rods - Specimens with built-in (BI) anchors from the manufacturer:									
BI#3-1	18,700 (129)	18,000 (124)	1.04	30.8 (137.5)	34.7 (154)	280 (1,929)	315 (2,172)	0.89	S
BI#3-2	18,900 (130)	18,000 (124)	1.05	30.9 (138.0)	34.7 (154)	281 (1,936)	315 (2,172)	0.89	RH
BI#3-3	19,800 (136)	18,000 (124)	1.10	32.1 (143.4)	34.7 (154)	292 (2,012)	315 (2,172)	0.93	SH
BI#3-4	18,900 (130)	18,000 (124)	1.05	29.6 (132.1)	34.7 (154)	269 (1,854)	315 (2,172)	0.85	S
#3 (10 mm) CFRP rods - Specimens with bond-type (BT) anchors fabricated in the laboratory:									
BT#3-1	19,500 (134)	18,000 (124)	1.08	28.1 (125.4)	34.7 (154)	255 (1,760)	315 (2,172)	0.81	S
BT#3-2	20,500 (141)	18,000 (124)	1.14	30.0 (134.2)	34.7 (154)	273 (1,883)	315 (2,172)	0.87	S
BT#3-3	18,000 (124)	18,000 (124)	1.00	39.2 (174.9)	34.7 (154)	356 (2,455)	315 (2,172)	1.13	S
BT#3-4	19,000 (131)	18,000 (124)	1.06	35.8 (159.6)	34.7 (154)	325 (2,239)	315 (2,172)	1.03	RH
BT#3-5	19,500 (134)	18,000 (124)	1.08	39.4 (176.0)	34.7 (154)	358 (2,470)	315 (2,172)	1.14	S
BT#3-6	20,000 (138)	18,000 (124)	1.11	31.0 (138.7)	34.7 (154)	282 (1,946)	315 (2,172)	0.90	SH
#4 (13 mm) CFRP rods - Specimens with built-in (BI) anchors from the manufacturer:									
BI#4-1	18,800 (130)	18,000 (124)	1.04	67.2 (299.6)	58.8 (262)	343 (2,365)	300 (2,068)	1.14	S
BI#4-2	19,300 (133)	18,000 (124)	1.07	65.3 (290.7)	58.8 (262)	333 (2,295)	300 (2,068)	1.11	R
BI#4-3	20,300 (140)	18,000 (124)	1.13	60.6 (270.1)	58.8 (262)	309 (2,132)	300 (2,068)	1.03	R
BI#4-4	19,200 (133)	18,000 (124)	1.07	49.0 (218.6)	58.8 (262)	250 (1,726)	300 (2,068)	0.83	S
#4 (13 mm) CFRP rods - Specimens with bond-type (BT) anchors fabricated in the laboratory:									
BT#4-1	21,000 (145)	18,000 (124)	1.17	61.3 (273.5)	58.8 (262)	313 (2,159)	300 (2,068)	1.04	RH
BT#4-2	19,500 (134)	18,000 (124)	1.08	68.8 (306.4)	58.8 (262)	351 (2,419)	300 (2,068)	1.17	S
BT#4-3	20,000 (138)	18,000 (124)	1.11	76.2 (340.1)	58.8 (262)	389 (2,685)	300 (2,068)	1.30	R
BT#4-4	19,000 (131)	18,000 (124)	1.06	62.5 (278.8)	58.8 (262)	319 (2,201)	300 (2,068)	1.06	R
BT#4-5	20,500 (141)	18,000 (124)	1.14	70.0 (312.0)	58.8 (262)	357 (2,463)	300 (2,068)	1.19	R
BT#4-6	Used for cyclic and sustained loading tests								

Note (1): E_{measured} is the measured modulus of elasticity, $E_{\text{manufacturer}}$ is the design modulus of elasticity reported by the manufacturer, P_{max} is the measured maximum tensile load, $P_{\text{manufacturer}}$ is the ultimate tensile load reported by the manufacturer, F_{max} is the measured maximum tensile load divided by the nominal cross-sectional area of CFRP rods, $F_{\text{manufacturer}}$ is the guaranteed tensile strength reported by the manufacturer.

Failure modes: slippage (S), rupture of helical CFRP (RH), CFRP shear fracture at the anchorage edge (SH), rupture of CFRP rod (R).

Note (2): Two different test setups were used for BI and BT specimens (see Figs. 1 and 5)

stress was higher when using 12 in. (305 mm) anchor. Even though both anchor lengths did achieve the guaranteed capacity of #4 (13 mm) rods, the failure mode shifted from failure at the anchor edge for 12 in. (305 mm) anchors to CFRP rupture at the free length for 15-in. (381-mm) anchors (Fig. 8). This indicated that the bond stress was more evenly distributed when using longer bond length. If it is desirable to (conservatively) keep the average bond stress less than 2.5 ksi (17 MPa), as was found based on the results of #3 (10 mm) CFRP anchors, the minimum required length for #4 (13 mm) anchors would be 15 in. (381 mm). It can be observed in Fig. 13 that the 12-in. (305-mm) anchors could achieve the maximum guaranteed strength. This generates an average bond stress of 3.12 ksi (22 MPa). On the other hand, the maximum guaranteed strength will be achieved by only 2.49 ksi (17 MPa) bond stress when using 15-in. (381-mm) anchors. The failure loads of specimens BT#4-3 and BT#4-5 were 130% and 119% of maximum guaranteed strength, respectively. The average is 124.5%. This failure load causes an average bond stress of 3.1 ksi (21 MPa). It can be concluded that the maximum average bond stresses before failure in order to develop the guaranteed capacities of CFRP for #3

(10 mm) and #4 (13 mm) rods are 2.5 ksi (17 MPa) and 3.0 ksi (21 MPa), respectively. It can also be observed from Fig. 13 that the anchor stiffness appeared to be proportional to the bond length. Longer anchors resulted in higher anchorage stiffness as shown in the load vs slip curves, and as it was observed previously [20,23]. Finally, it was observed that increasing the bond length delayed the slip initiation. For #3 (10 mm) anchors, the slip was initiated when the load reached about 10% the guaranteed strength for 10-in. (254-mm) anchors, while it was initiated when the load reached 30% or more for 12-in. (305-mm) anchors. In the case of #4 (13 mm) anchors, the slip was initiated almost immediately for 12-in. (305-mm) anchors, while slip was initiated when the load reached more than 30% for 15-in. (381-mm) anchors.

3.5. Effects of cross-sectional area ratio

The results of specimens BT#3-3, BT#3-4, BT#3-5, BT#4-1, BT#4-2, and BT#4-4 can be compared to investigate the effects of cross-sectional area ratio of CFRP rods to the anchor's borehole. The anchor length was equal to 12 in. (305 mm) for all these spec-

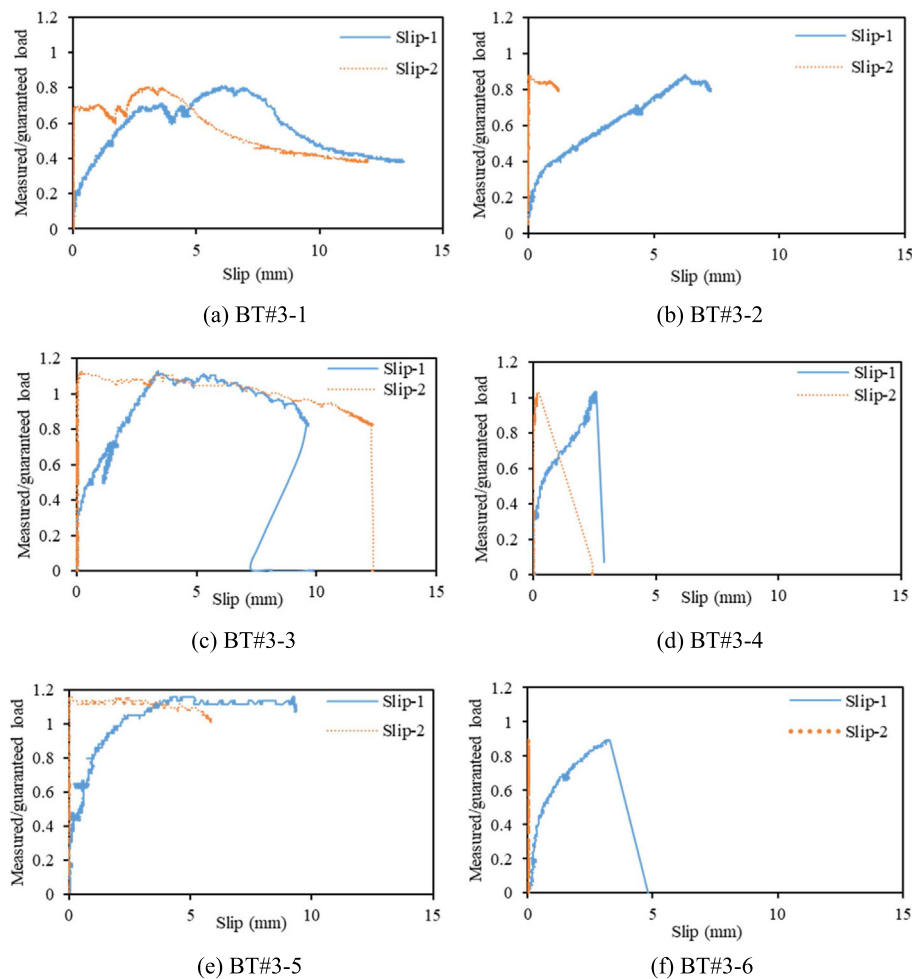


Fig. 11. Load-slip relationships for #3 (10 mm) CFRP rods tested with proposed bond-type anchors (1.0 in. = 25.4 mm).

imens. The only variable was the size of CFRP rods. This led to different cross-sectional area ratios; 22.7% and 40.4% for #3 (10 mm) and #4 (13 mm) CFRP rods, respectively. It can be observed from Fig. 14 that the specimens with higher cross-sectional area ratios (specimens BT#4-1, BT#4-2, and BT#4-4) achieved higher tensile loads. That was also true for the average bond stress. The maximum tensile load for #3 (10 mm) anchors resulted in an average bond stress equal to 2.71 ksi (19 MPa), whereas the maximum load for the #4 (13 mm) anchors generated an average bond stress of 3.34 ksi (23 MPa). Since all the specimens compared herein had the same bond length, it can be stated that the tensile capacity and stiffness of CFRP anchors can be improved by increasing the cross-sectional ratio of CFRP rod to borehole. This conclusion is limited to 12 in. (305 mm) anchors and to the ratios tested in this study. The results of the present study agree with the results in the literature [42,59,60]. Smaller cross-sectional area ratio of rod to borehole produces larger grout thickness, which increases deformation and decreases tensile capacity.

3.6. Effects of curing time of the expansive grout

Only three samples were compared to examine the effects of grout's curing time. Anchors BT#3-1 and BT#3-2 had a curing time of four days. However, the curing time for anchor BT#3-6 was four weeks. The same loading procedure was followed for all specimens. The results indicated that the tensile strength of specimens with a curing time of four days only (BT#3-1 and BT#3-2) was only

6% less than the tensile strength of specimen BT#3-6, which was cured for four weeks (Fig. 15). However, after 20% loading, the anchorage stiffness was much less for the specimens with four-day curing time. As mentioned earlier, the anchorage stiffness refers to the slope of the load-slip relationship curve. This significant difference in stiffness could be explained by observing that the mode of failure for these three specimens was excessive slip-page due to interior surface shear at the grout itself. Since specimens with #3 (10 mm) CFRP rods had a cross-sectional area ratio of CFRP rod to anchor's borehole of 22.7%, there were a thick layer of the cementitious grout around the CFRP rod. Since the control failure mode for these specimens was directly proportioned to the grout material, curing time significantly affected the anchorage stiffness. However, the difference in tensile strength is not significant. The effects of curing time when the cross-sectional area ratio is different has not been further investigated. Moreover, the results reported in this study regarding the effects of curing time are based on testing three specimens only. Therefore, more tests are required to confirm the results associated with curing time effects of the behavior of BT anchors.

3.7. Cyclic and sustained loading

Specimen BT#4-6 was evaluated under cyclic and sustained loadings. The objective was to determine the loss in prestressing load and compare it with the acceptance criteria by the British Standards Institution [61]. The load was increased in stages to

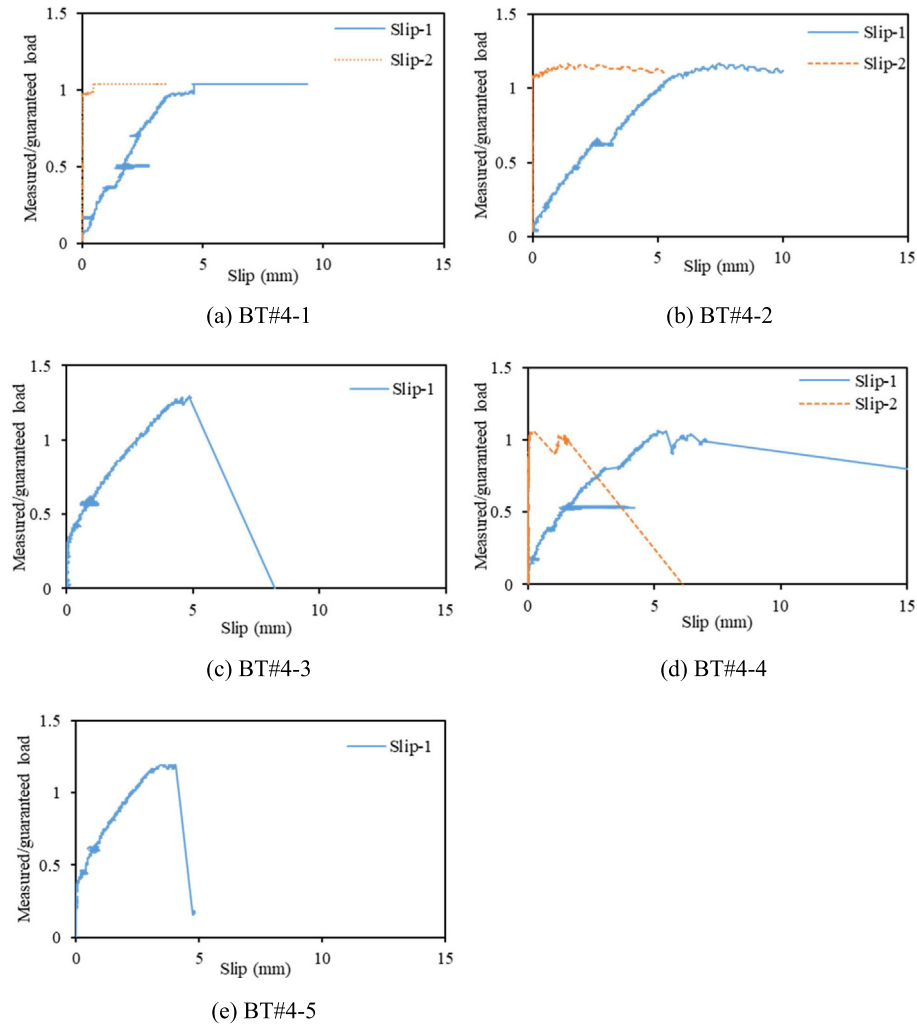


Fig. 12. Load-slip relationships for #4 (13 mm) CFRP rods tested with proposed bond-type anchors (1.0 in. = 25.4 mm).

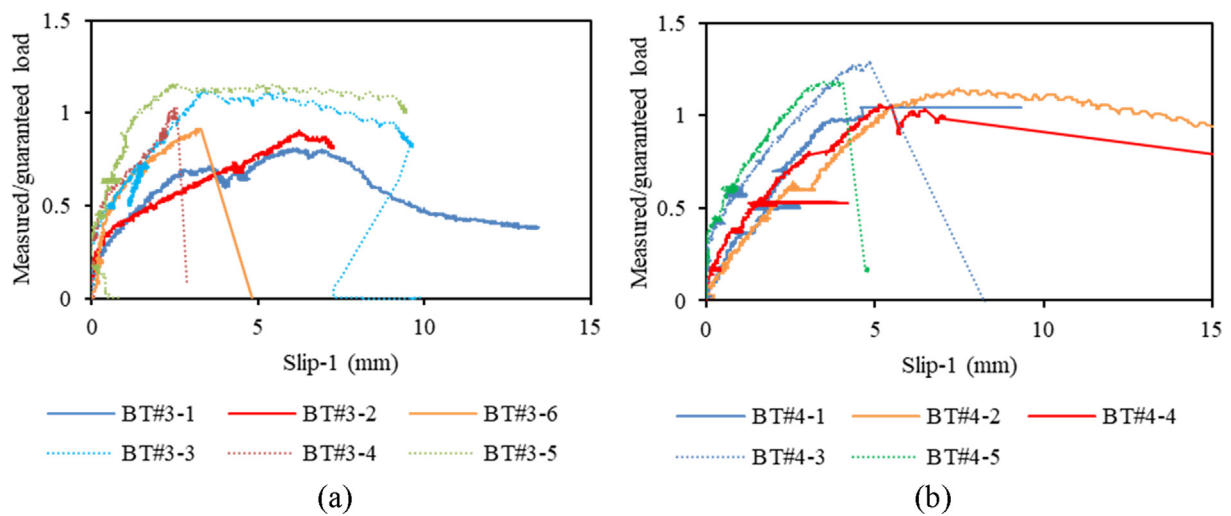


Fig. 13. Effects of bond length; (a): #3 (10 mm) CFRP anchors, and (b): #4 (13 mm) CFRP anchors.

10%, 20%, 30, 40%, 50%, and 60% of the guaranteed tensile capacity (F_u), and then increased and locked at 70% of F_u to evaluate the behavior of the anchor under sustained loading. A prestressing force of $0.7F_u$ was slightly higher than what ACI 440.4R [62] recommends as the maximum jacking force ($0.65F_u$). Fig. 16(a) shows the

load vs slip cyclic behavior of the anchor. The first two cycles did not result in noticeable slippage. Cycles three through six are labeled in the figure for clarity. Permanent displacement of 0.7 mm was recorded after the sixth cycle. Then, the load was increased and locked at $0.7F_u$ so that the evaluation of the sus-

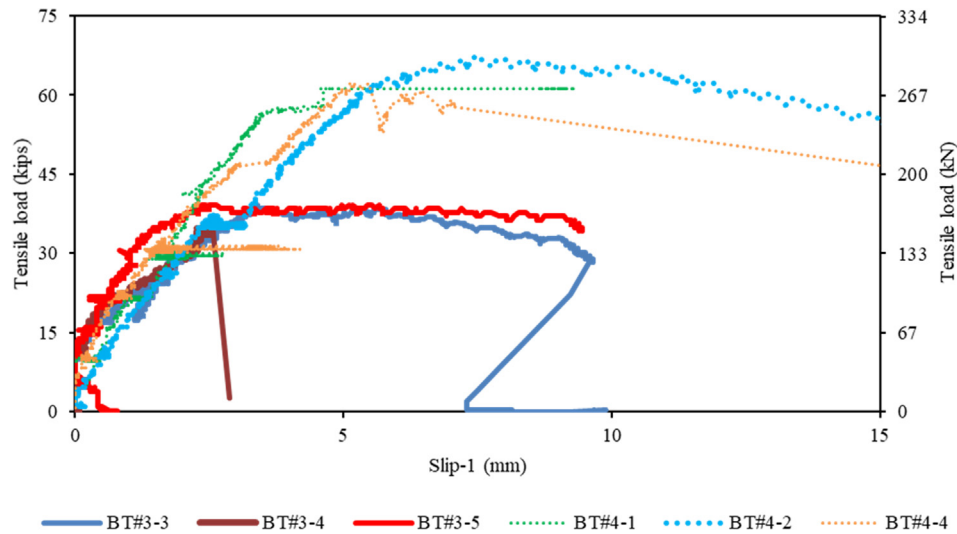


Fig. 14. Load vs slip relationship showing the effects of cross-sectional area ratio of CFRP rod to the borehole of the anchor.

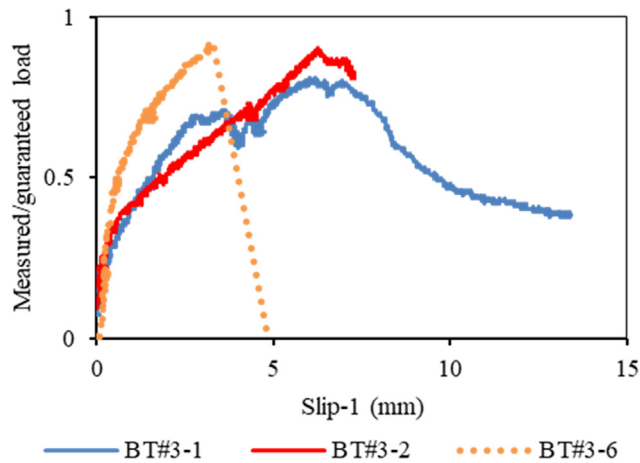


Fig. 15. Load vs slip relationship showing the effects of grout curing time; four days for specimens 3-1 and 3-2, and four weeks for specimen 3-6.

ment recorded by the “LVDT readings” minus the elastic elongation of the CFRP rod (labeled in Fig. 16b as “Elas. Elong”).

The losses in prestressing force were monitored for ten days to be compared with BSI criteria, as shown in Fig. 17(a). The losses exceeded the acceptance criteria. However, the losses rate or the rate of decrease in prestressing force stopped after the seventh day, at which the losses were 10%. Therefore, initial losses of at least 10% must be considered in the design procedure if this anchor is used in practice. The losses, slippage, and the load as a percentage of the guaranteed tensile capacity are presented in Fig. 17(b). It can be noticed that 79% of the losses occurred in the 1st day. The force dropped from 70% F_u to 64% F_u after one day, and it dropped to 62% F_u after ten days. The slippage increased rapidly in the first day, but it was fluctuating up and down after that due to the increase of elastic elongation, creep, and relaxation of the anchorage system. It can be concluded from Fig. 17 that even though the losses exceeded BSI acceptance limits, the results are promising because the losses stopped after seven days. Of course, other types of anchorage systems are available in the market, but the aim of this study was to provide another alternative and an additional data point that can be used in the future to develop the desired anchorage system. It should be mentioned that the results of the cyclic and sustained loading response are based on testing one specimen only. Therefore, future researchers are encouraged to conduct more tests to confirm the results.

tained loading behavior could be started immediately after the cyclic loading test. The slippage and elastic elongation are presented in Fig. 16(b). The “Net Slip” is basically equal to the total displace-

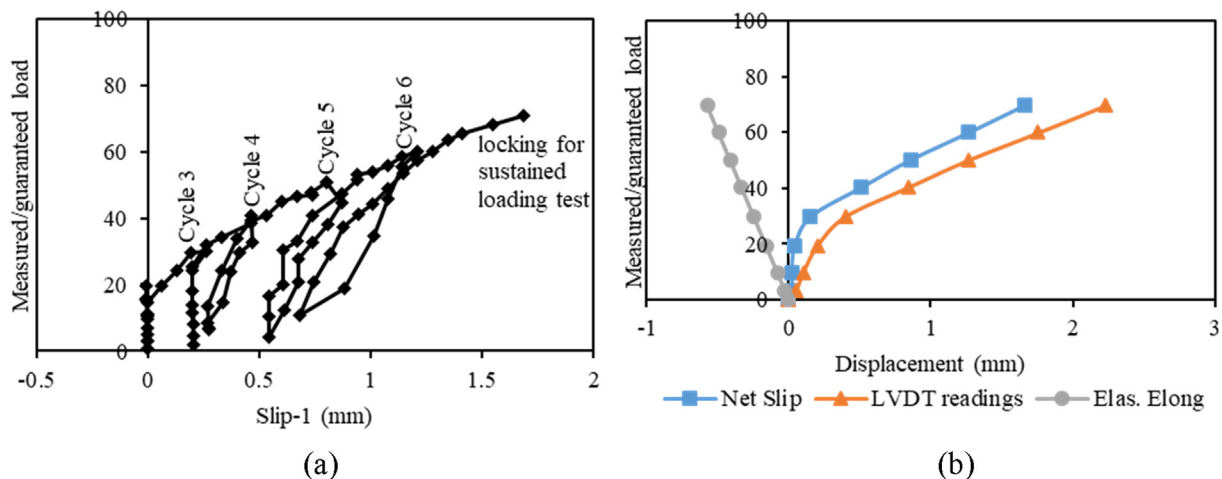


Fig. 16. Cyclic tensile behavior: (a) load vs slip relationship; (b) load vs displacement/ slip/ elastic elongation relationships.

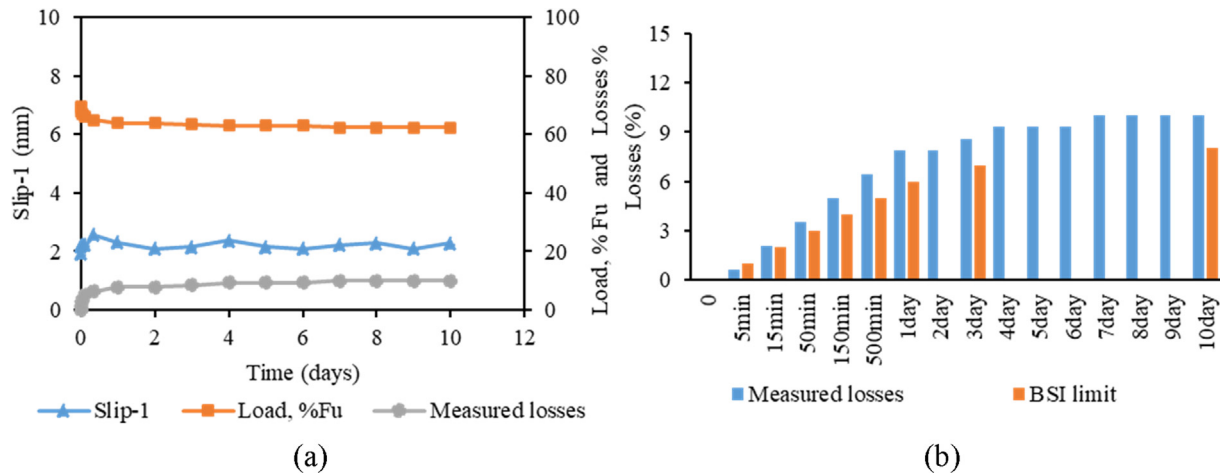


Fig. 17. Sustained loading behavior: (a) Measured losses vs BSI criteria; (b) Changes in slip, losses, and prestressing force over time.

4. Numerical modeling

4.1. FE modeling and material properties

Finite element (FE) analysis for the tested specimens was conducted using a finite-element software commercially known as ABAQUS [63]. 3D deformable parts were created for steel tube, grout, and CFRP rods as shown in Fig. 18. All parts were modeled as solid-homogeneous sections and meshed with hexahedral, eight-node linear brick elements with reduced integration. CFRP rods were modeled as an elastic material with a modulus of elasticity determined from the experimental tests (Table 4) and a Poisson's ratio of 0.25 [64]. CFRP rods considered failed by rupture and the analysis stopped when CFRP tensile stress reached the ultimate strength determined from the experimental tests. It should be noted that the measured engineering stresses from the experimental tensile tests were converted to true stresses in the FE analysis.

Because the steel tube did not yield in any of the tested specimens and is not expected to yield in the FE analysis (confirmed later by the FE model), steel tube was also modeled as elastic material with a typical modulus of elasticity and Poisson's ratio of 200 GPa and 0.3, respectively. As stated in previous research [65], it is difficult to test the compressive and tensile properties of the expansive grout. Therefore, it was simulated as an elastic material with an established modulus of elasticity of 42.5 GPa [65] and a Poisson's ratio of 0.19 reported by Hyett et al. [66]. The plastic deformation of the grout was considered when modeling the inter-

action between the grout and CFRP rods, as explained in section 4.2.

4.2. Bond properties

Perfect bond was considered between the grout and the interior surface of the steel tube. The two surfaces were constrained using tie constraint distributed along the bond length. Standard surface-to-surface contact was assumed between the inside surface of the grout and the outside surface of CFRP rods, with tangential friction coefficient of 0.39 as reported by Schesser et al. [65]. In addition, the two surfaces were connected using nonlinear axial springs (Fig. 19), the axial stiffness of which was estimated depending on the empirical bond-slip relationship observed in this experimental study. Fig. 20 shows the assumed empirical bond-slip relationships for #3 (10 mm) and #4 (13 mm) rods that were used to input the axial stiffness of the springs in the FE model.

4.3. Boundary conditions, loading and analysis

Encastre boundary condition (no translation and no rotation in all directions) was assigned to the external surface of the steel tube at the same location where the hex nut was placed in the experimental setup (Fig. 6). Displacement-control loading was applied at the tip of CFRP rod. A maximum applied displacement of 8 mm was considered. Static analyses were performed with an initial increment size of 0.01 mm and a maximum of 0.1 mm.

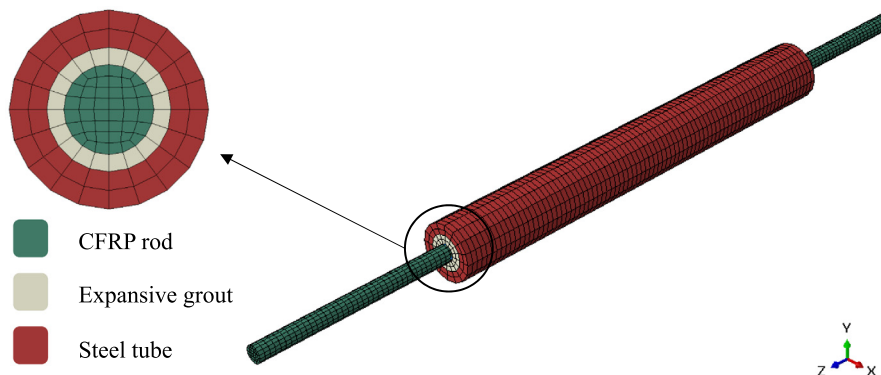


Fig. 18. 3D finite element model.

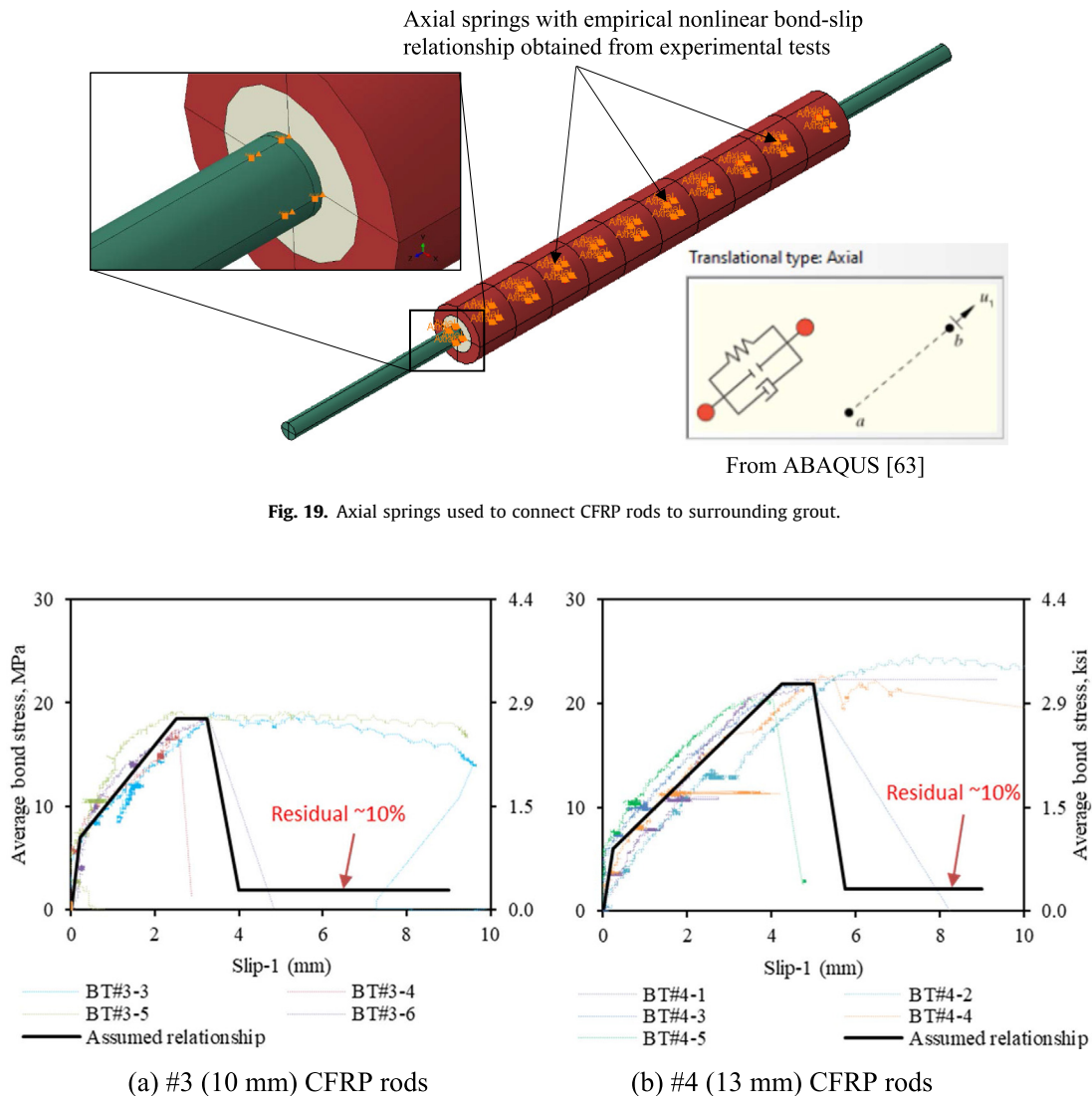


Fig. 20. Empirical bond-slip relationship based on experimental results.

4.4. Numerical-experimental comparison

The predicted load-slip curves are compared to the experimental results in this section. In the numerical model, slip-1 was extracted from a predefined monitoring node on the surface of CFRP rod located at the vicinity of the anchor loading end (Fig. 6). Fig. 21 compares the load-slip curves obtained from the FE model with those obtained from experimental tests. It can be noted in the graphs in Fig. 21 that the FE results are in good agreement with the experimental results. The FE model well predicted the tensile behavior of the BT anchors in terms of load-slip stiffness, ultimate load, and slippage. Because the bond between the grout and steel tube was assumed to be perfect, the predicted load-slip relationship is expected to be slightly stiffer than the that obtained from the experimental results. The maximum predicted loads from the FE model and the maximum achieved loads from the experimental tests are compared in Table 5. Except for specimen BT#4-5, the FE model slightly underestimated the pullout strength of the BT anchors. The predicted failure modes were excessive slippage (S) for the first three models, but it was CFRP rupture (R) for #4 (13 mm) CFRP rods with 15 in. (381 mm) BT anchors.

4.5. Parametric study – Bond length

Using the verified FE mode, a parametric study to investigate the sensitivity of the pullout strength to the change of anchor bond length was conducted. The bond length varied from 6 in. (152 mm) to 18 in. (457 mm). Fig. 22 shows how the predicted pullout strength changes with the change of bond length. As expected, the pullout strength increased by increasing the bond length for both sizes of CFRP rods. Based on the FE model, it seems that a bond length of 15 in. (381 mm) is required to ensure CFRP rupture instead of slippage failure for #3 (10 mm) and #4 (13 mm) CFRP rods. Finally, Fig. 22 also shows that anchors with #4 (13 mm) CFRP rods achieved higher loads than anchors with #3 (10 mm) rods for the same bond length (anchor length). This indicates that higher ratio of rod diameter to anchor hole diameter results in higher strength, as observed in the experimental tests.

5. Anchorage applications

First, the proposed anchor can be used to test FRP rods under tensile loading, especially for research purposes. The anchor is suitable whenever grabbing FRP rods is required. The same anchorage

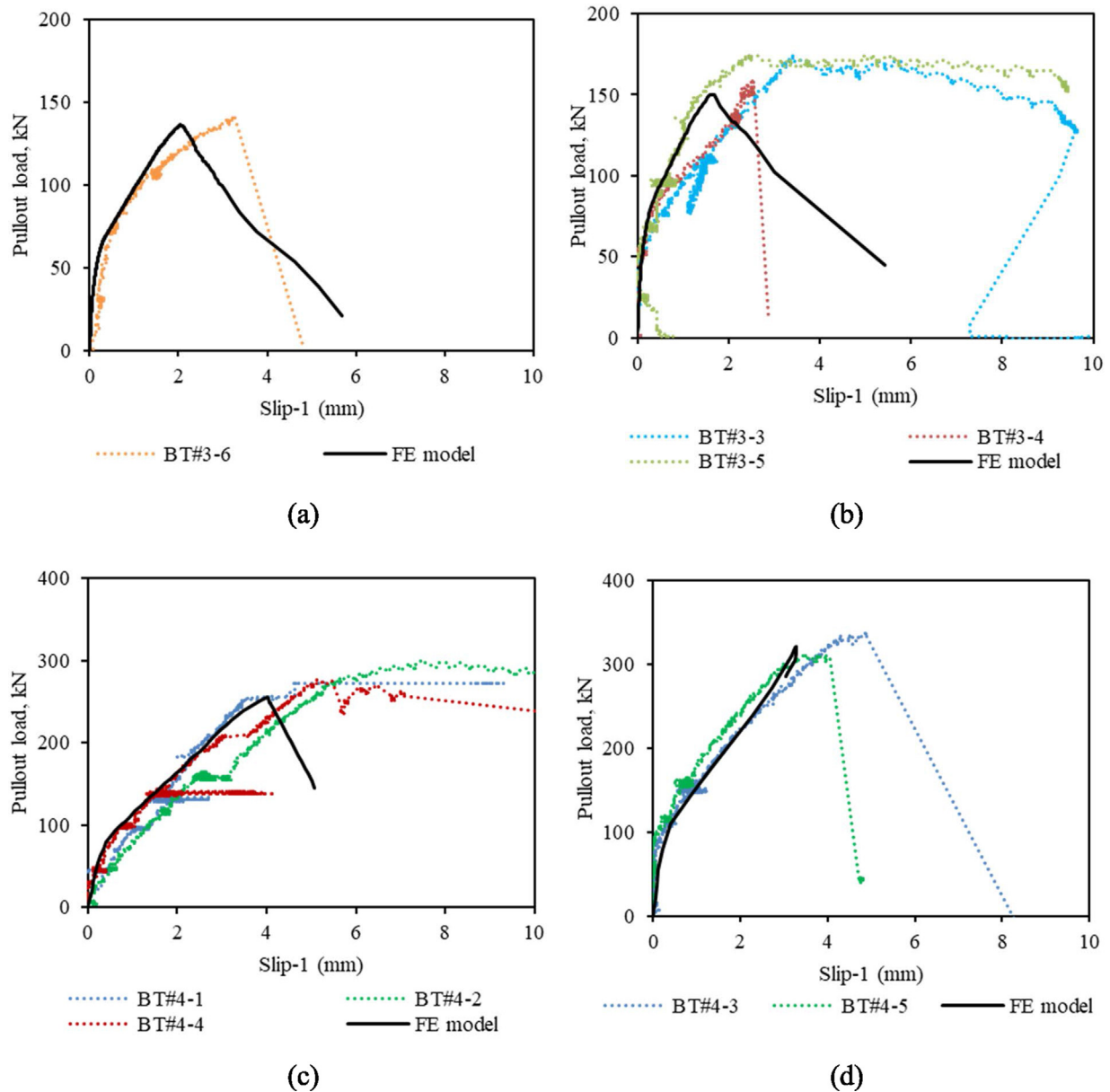


Fig. 21. Numerical vs Experimental load-slip curves: (a) anchor length = 10 in. (254 mm), (b) anchor length = 12 in. (305 mm), (c) anchor length = 12 in. (305 mm), (d) anchor length = 15 in. (381 mm).

Table 5

Numerical vs experimental results.

Specimen	Rod diameter, in. (mm)	Anchor length, in. (mm)	Experimental		Numerical		Numerical/ experimental
			P_{max} , kip (kN)	Failure mode	P_{max} , kip (kN)	Failure mode	
BT#3-6	0.375 (10)	10 (254)	31.0 (138.7)	SH	30.7 (136.3)	S	0.99
BT#3-3	0.375 (10)	12 (305)	39.2 (174.9)	S	33.7 (150.0)	S	0.86
BT#3-4			35.8 (159.6)	RH			0.94
BT#3-5			39.4 (176.0)	S			0.86
BT#4-1	0.500 (13)	12 (305)	67.2 (299.6)	S	57.5 (255.6)	S	0.86
BT#4-2			65.3 (290.7)	R			0.88
BT#4-4			62.5 (278.8)	R			0.92
BT#4-3	0.500 (13)	15 (381)	76.2 (340.1)	R	72.2 (321.0)	R	0.95
BT#4-5			70.0 (312.0)	R			1.03

Failure modes: slippage (S), rupture of helical CFRP (RH), CFRP shear fracture at the anchorage edge (SH), rupture of CFRP rod (R).

system was used to perform pullout tests on CFRP rods [67,68]. Secondly, the new bond-type anchorage system may be considered in FRP ground anchor applications. As mentioned before, the

threaded steel tube that was used to fabricate the proposed anchor in this study is originally a geo-drill injection anchor system for soil-anchorage applications. Steel strands and wires are the com-

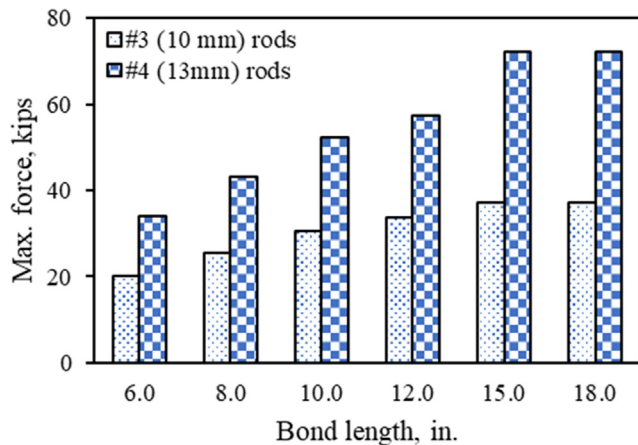


Fig. 22. Parametric study, maximum force vs anchorage bond length. Units; 1.0 in. = 25.4 mm, 1.0 kip = 4.448 kN.

mon materials that are used in ground anchor applications. However, the use of FRP ground anchors has become a suitable alternative in certain aggressive environments [7,12]. Therefore, the proposed CFRP anchor may be an appropriate alternative to replace the available steel ground anchors when necessary. Furthermore, FRP materials have been suggested to be considered as a replacement for steel wires in cable structures [2,6]. The proposed anchor may be considered in cable structures given the fact that it consisted of a threaded tube with a nut that can easily be locked and released. Of course, further investigations need to be performed before the anchor is used in practice due to the limited tests performed in this study. Finally, the proposed BT anchor may potentially be used in prestressed concrete applications [18,69,70]. For example, Saeed and Rad [71] used the proposed anchor to solve the bond problems associated with CFRP pre-tensioned concrete beams.

6. Conclusions & observations

The lack of information regarding CFRP bond-type anchors using an expansive grout as a filling material motivated the authors to conduct this experimental investigation. A total of twelve bond-type anchors and eight CFRP rods with pre-attached steel anchors from the manufacturer were tested and evaluated in this study. Pullout tests were conducted on #3 (10 mm) and #4 (13 mm) CFRP rods. Short-term and long-term tests were performed to evaluate a proposed bond -type anchorage system. The following conclusions are made based on the results of this study:

- For #3 (10 mm) CFRP rods, only the 12 in. (305 mm) bond-type anchors achieved the guaranteed tensile capacity. However, all bond-type anchors for #4 (13 mm) CFRP rods, 12-in. (305 mm) and 15 in. (381 mm), were able to achieve the guaranteed tensile strength. The measured tensile strength for #4 (13 mm) CFRP rods ranged from 4% to 30% higher than the guaranteed tensile capacity.
- Determining the anchor length by assuming a maximum average bond stress of 2.50 ksi (17 MPa) appears to be a safe approach based on the results of this study. Based on this method, bond lengths of 12 in. (305 mm) and 15 in. (381 mm) are required for #3 (10 mm) and #4 (13 mm) CFRP rods, respectively. Therefore, the recommended minimum bond length is 32 times rod diameter for #3 (10 mm) rods and 30 times rod diameter for #4 (13 mm) rods.

- Increasing the anchorage bond length delayed the slippage initiation and increased the anchorage stiffness.
- Larger cross-sectional area ratio of CFRP rod to borehole limited the grout deformation, increased the load-slip stiffness, and consequently increased the tensile strength. This conclusion is only true for the two ratios used in this study. More tests are required to generalize this statement.
- The tensile capacity of #3 (10 mm) CFRP anchors with a curing time of four days was 6% less than that of specimens with four weeks of curing. This observation is based on testing three specimens only.
- Based on the cyclic and sustained loading tests, the losses of the proposed anchor exceeded BSI limits but stopped almost completely after seven days. This observation was made based on the data collected for ten days and based on testing a single specimen. More tests are required to confirm this conclusion.
- The numerical FE model well predicted the tensile strength and load-slip stiffness of the tested specimens.
- Based on the FE model, a bond length of 15 in. (381 mm) is required for guaranteed rupture failure of #3 (10 mm) and #4 (13 mm) CFRP rods.

CRediT authorship contribution statement

Yasir M. Saeed: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Salam M. Al-Obaidi:** Investigation, Writing - review & editing. **Ehab G. Al-hasany:** Software. **Franz N. Rad:** Validation, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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